

Hariom Pandey

AI Engineer & Backend Developer | LLMs, RAG, Fine-Tuning

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PROFESSIONAL SUMMARY

AI Engineer specializing in LLM fine-tuning, Retrieval-Augmented Generation, and agentic AI workflows, with a solid backend and DSA foundation (400+ problems solved). Focused on building scalable, intelligent, real-world AI systems.

EDUCATION

Indian Institute Of Technology Madras

Dec 2026

BS in Data Science and Applications (Minor: AI, Economics & Finance)

CGPA: 7.5

WORK EXPERIENCE

INFINITY LEARN

Sep 2025 – Mar 2026

Data Analytics & ML Intern

- Analyzed large-scale student data using SQL and Python, delivering actionable insights on learning trends and behavioral patterns to stakeholders.
- Architected and deployed interactive Metabase dashboards on Snowflake, tracking key metrics across thousands of students and accelerating operational decision-making.
- Designed and deployed a Student Attendance Prediction system using XGBoost and ensemble methods, achieving 80% accuracy through feature engineering and cross-validation.
- Collaborated with cross-functional teams to translate business requirements into SQL queries and Snowflake data pipelines, improving reporting turnaround time.

ACADEMIC PROJECTS

Sportsclub App – RAG Chatbot + AI-Generated PDF Reports

2026

Tech: React, FastAPI, PostgreSQL, Gemini API, Pinecone, ReportLab, Vercel, Render, Neon

- Independently designed and built a full-stack sports club management app end-to-end – React frontend, FastAPI backend, and PostgreSQL schema for members, stats, and schedules.
- Built a RAG-based chatbot using Pinecone for semantic retrieval over club data, feeding grounded context to the Gemini API for accurate, hallucination-reduced responses.
- Automated LLM output into structured, downloadable PDF reports using ReportLab, replacing manual reporting.
- Implemented pytest unit/API tests, automated via a GitHub Actions CI/CD pipeline, and deployed on Vercel, Render, and Neon.

Automated PR Reviewer Bot

2026

Tech: Python, GitHub Webhooks, GitHub Actions, Gemini API

- Built an automated code review bot that listens to GitHub PR webhook events, parses the diff, and posts LLM-generated review comments directly on the pull request.
- Integrated with GitHub Actions to trigger review runs automatically as part of the CI/CD pipeline on every PR push.
- Engineered prompts to target code quality, bugs, and style, reducing manual first-pass review effort.

Legal & Policy Document Classifier (Transformer + PEFT)

2026

Tech: Python, HuggingFace Transformers, LoRA/QLoRA (PEFT), legal-bert-base-uncased

- Built a Transformer-based text classifier for legal and policy documents in a college-hosted Kaggle competition, evaluated on accuracy.
- Applied LoRA (PEFT) on legal-bert-base-uncased with head+tail truncation and weighted loss to handle long-document and class-imbalance challenges.
- Iterated from a DistilBERT baseline to domain-specific LegalBERT + LoRA, improving accuracy from 0.35 to 0.69 (~97% relative gain).

CERTIFICATIONS & ACHIEVEMENTS

- Data Structures and Algorithms – GeeksforGeeks
- Python and SQL (Advanced) – HackerRank
- Achieved Top 20 rank in a Kaggle competition on System Threat Forecasting (prediction-based challenge).
- Solved 500+ DSA problems across GeeksforGeeks, LeetCode, and HackerRank.
- Built a fine-tuned Wav2Vec2-XLS-R model for Uyghur speech-to-text (ASR) in a college Kaggle competition, achieving 5.8% CER.

SKILLS

Machine Learning: Supervised Learning, Unsupervised Learning, Ensemble Methods, Model Optimization, XGBoost

Deep Learning: Neural Networks, Backpropagation, Optimization Algorithms, Transformer Architectures, PyTorch

Gen AI: LLMs, RAG, NLP, Transformers, ASR, Speech Processing, CV, Gemini API, LangChain, LangGraph, HuggingFace Transformers, Fine-Tuning (LoRA/QLoRA), Prompt Engineering, Embeddings, Scikit-learn

Programming, Cloud & Deployment: Python, SQL, Java, Snowflake, FastAPI, REST APIs, Vercel, Render

Soft Skills: Communication, Teamwork, Problem Solving, Time Management, Adaptability